Wood Glass Googless

Material state. Observable views.
Behavior assembled from LUTs.



A formalization of the paradigm of

Tom Klootwijk

A deterministic, bounded runtime maps validated state to versioned behavior descriptors, assembles a compatible component graph, and publishes a coherent snapshot for mechanics, rendering and inspection. The initial physical example is a degrading wooden column; the architecture also supports other material and observation models.

VERSION 1.0 | 30 OCTOBER 2026

Technical specification with worked model, runtime contracts and references

01 / DEFINITION

Scope and reading guide

THE PARADIGM IN ONE PARAGRAPH

Core proposition. A family of typed lookup tables maps validated system state to compatible behaviors. The runtime forms the execution graph from those selections and publishes one immutable snapshot for calculation, observation and rendering.

This edition formalizes the supplied conversation [U1]. Its later instruction to retain the mechanics, exclude religious or philosophical framing, and keep personal identity out of the physical model governs the technical scope. Earlier generated explanations are treated as draft material to examine, not as measurements or established results.

Name	Meaning used in this edition	Status
Wood	The evolving material domain: geometry, mechanics, chemical inventories and heat.	Grounded in [U1].
Glass	The observable interface: optical models, rendered views and inspectable evidence.	Working editorial definition.
Googless	The user-supplied name of the integrated architectural paradigm. Its expansion is unspecified.	Name preserved; no provider policy inferred.

The Glass interpretation can be renamed without changing the formal machine. The source does not establish that Googless means Google-free or refers to a particular commercial product. External services, if later connected, enter through explicit input adapters.

Reading route

Pages	Purpose
3-6	Architecture, formal state, typed LUTs and transitions.
7-13	Wood mechanics, worked example, FEM, heat, observation, optics and log-polar encoding.
14-19	Runtime composition, concurrency, operators, determinism, resources and execution.
20-24	Conformance checks, corrections, notation and cited sources.

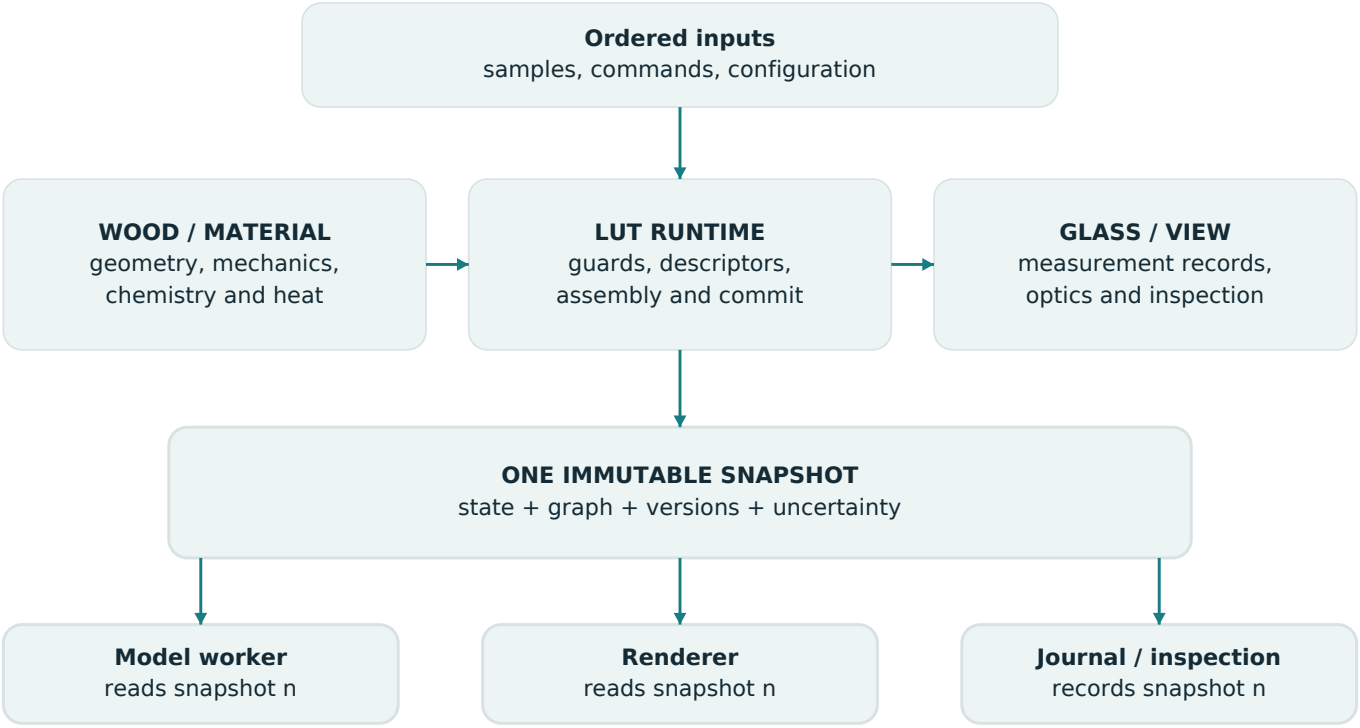
Attribution supplied by the requester: Tom Klootwijk; reference NL200678942; date of birth as supplied, 10-07-1990. These fields are document metadata only. No physical parameter, security credential or runtime key is derived from them.

Document status: version 1.0, an engineering specification and illustrative numerical model. Equations retain their stated assumptions; implementation conformance and physical calibration are separate activities.

02 / SYSTEM VIEW

One architecture, three planes

The framework separates the material being modeled, the machinery that evaluates it, and the information exposed to an observer. A state change can select new behavior without changing the meaning of a variable or invalidating an existing reader.



Plane	Owns	Produces
Material / Wood	Geometry, material laws, reaction and thermal inventories, numerical state.	A candidate physical state and model diagnostics.
Runtime	Input ordering, guards, LUT versions, behavior registry, graph assembly and publication.	A coherent, validated snapshot or explicit rejection.
Observation / Glass	Measured data with provenance; geometry and optical views; uncertainty display.	Evidence records and images tied to a snapshot version.

Boundary decisions

The computational core accepts only explicit events and configuration. It can be replayed without repeating live acquisition. A physically closed simulation has a separate mass-energy boundary, defined on page 10. Neither boundary makes the modeled world perfectly known.

The renderer consumes a committed snapshot. If an optical sensor is part of a later control loop, its output returns as a timestamped event, with delay and noise represented. A frame must never silently become ground-truth input to the same simulation step.

03 / MATHEMATICAL CONTRACT

Formal machine and invariants

Let C be the immutable configuration bundle for a run: typed LUTs L, approved behavior registry B, physical parameters Θ, numerical policy N and version manifest V. Define:

$$C = (L, B, \Theta, N, V)$$
$$S_k = (k, t_k, q_k, x_k, b_k, G_k, v_k)$$

Here k is the commit sequence, t simulated time, q the discrete material regime, x the modeled physical state, b the observation/uncertainty record, G the runtime component graph, and v the version manifest. Runtime health is tracked separately. Result is a tagged success/error union. Δt_k is a positive accepted substep.

$$\text{Step}_C(S_k, e_k, \Delta t_k) \to \text{Success}(S_{k+1}) \mid \text{Error}$$
$$x^* = F_{\Theta,N,Gk}(x_k, \text{Validate}(e_k), \Delta t_k)$$
$$b^* = \text{Update}(b_k, e_k, x^*, \text{observation_model})$$
$$q^* = \delta(q_k, x^*, e_k)$$
$$d^* = L_{\text{behavior}}[q^*, \text{capabilities}, \text{model_version}]$$
$$(x\dagger, b\dagger) = \text{Migrate}(x^*, b^*, d^*)$$
$$G^* = \text{Assemble}(d^*, B)$$
$$S_{k+1} = \text{Commit}(k+1, t_k+\Delta t_k, q^*, x\dagger, b\dagger, G^*, v)$$

The star denotes a candidate, not an externally visible value. If a regime boundary lies inside the step, the selected solver must locate the event, split or reject the step, and apply the declared state transfer. It must not integrate past its valid domain and merely change a display label afterward. Migration transfers physical and estimator state to the selected representation; it is identity if no transfer is required [D1].

Invariant	Required meaning
I1 · Typed state	Every physical quantity carries a unit, domain and validity status. Unknown is explicit.
I2 · Coherence	State, LUT bundle, selected behaviors and view inputs refer to one committed epoch.
I3 · Bounded execution	Graph size, tables, queued events and retained snapshots respect declared limits.
I4 · Valid publication	Only a complete, checked candidate becomes current. Rejections retain the prior snapshot.
I5 · Causal evidence	Every observation distinguishes measured, estimated, simulated and illustrative values.
I6 · Attribution independence	Changing author metadata leaves equations, coefficients and numerical results unchanged.

A total result function covers unknown table keys, invalid geometry, missing capabilities, nonfinite arithmetic and budget exhaustion. The error path is part of the architecture.

04 / DATA MODEL

State, events and LUT contracts

Record	Minimum fields and semantics
Physical state x	Geometry/occupancy; material ID; displacement and velocity; thermal state; species inventories; validity bounds.
Event e	Source ID, source sequence, capture time, arrival time, accepted order, payload, units, provenance and quality flags.
Observation b	Measured or inferred quantity, uncertainty representation, measurement model, timestamp and calibration version.
Snapshot S	Run ID, commit epoch, simulated time, regime, state, graph, table manifest and diagnostic record.
Descriptor d	Behavior IDs, required interfaces, typed ports, parameter references, migration rule and resource ceilings.

Separate tables for separate meanings

LUT family	Key → value	Rule
Transition	Regime + event class → ordered guards and target rule	Guards are exhaustive or end in an explicit error.
Behavior	Regime + capabilities + model version → component descriptors	Values are approved IDs, not arbitrary executable addresses.
Material	Material ID + valid coordinates → constitutive/optical parameters	Calibration domain and units travel with the data.
Sampling	Quantized spatial/parameter coordinates → numeric field values	Interpolation and approximation error are declared.
Interface	Behavior ID → input/output contract and limits	Assembly rejects incompatible ports or missing dependencies.

TABLE CONTRACT

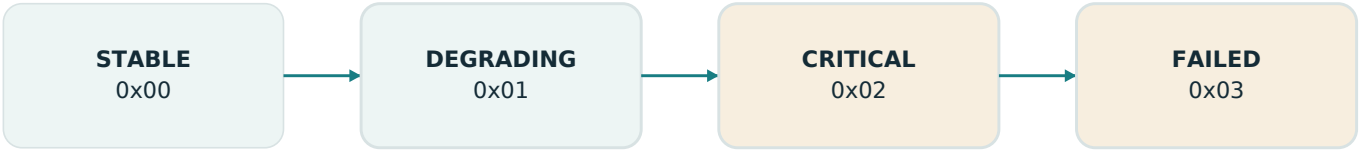
Every LUT bundle declares a schema version, dimensions, index bounds, units, provenance, interpolation policy, out-of-domain behavior and content digest. A digest detects content differences; authenticity requires an independently trusted source or verification mechanism.

A reproducible physical example uses a generated run identifier. Source content hashes and model versions identify artifacts. They do not convert a person’s identity into a modulus, geometric field, authentication proof or physical measurement.

05 / DISCRETE CONTROL

The transition system

Keep the four physical regimes requested in [U1], while distinguishing instability from confirmed fracture. This example uses an irreversible severity latch within one run. A reset creates a new initialized run; a reversible material model requires its own transition table.



Increasing severity within one run. Criticality can be reached directly; failure requires a declared failure event or law.

Priority	Guard / condition	Outcome
0	Input invalid, model outside domain, required behavior missing, or budget exceeded.	Reject candidate; pause or fault the runtime by policy. Preserve last valid material state.
1	Current regime is FAILED, or a validated failure event / configured failure law fires.	FAILED (0x03). Only a supported debris model may continue physical evolution.
2	Current regime is CRITICAL, or stability threshold is reached / cannot be ruled out.	CRITICAL (0x02). Localize the event; transfer to a valid critical model or pause.
3	Current regime is DEGRADING, or configured material-loss/change criterion is met.	DEGRADING (0x01). Continue only within the chosen model’s validity domain.
4	All earlier guards are false.	STABLE (0x00). This is a model regime, not a safety certification.

A concrete threshold

$$m = P_{cr}/P, \text{ with } P > 0$$
$$\text{critical_guard} = (\text{lower_bound}(m) \le 1 + \varepsilon_{guard})$$

Here m is a dimensionless model margin; ε_{guard} is a documented numerical or operational guard tolerance. With exact example parameters, $\varepsilon_{guard} = 0$. With uncertain geometry or modulus, use a declared valid bound-propagation method and report the reason for a critical classification. Missing/invalid bounds follow an explicit error or conservative-critical policy. $P \le 0$ needs a separate branch or rejection.

The stability test does not supply a fracture law. Failure of a linear equilibrium branch does not imply infinite velocity, zero mass or zero material stiffness.

06 / PHYSICAL MODEL

Wood: material and structural state

Use an ideal concentric cylindrical column as the first model. Outer radius R, inner radius r, length L, modulus E and density ρ are independent physical parameters. The surviving section is homogeneous in this reduced model. Require R, L, E, $K_{eff} > 0$, $\rho \geq 0$ and $0 \leq r < R$.

$$A = \pi(R^2 - r^2)$$
$$I = (\pi/4)(R^4 - r^4)$$
$$\mu = \rho A;\ B = EI$$

A: area [m²]; I: second moment of area [m⁴]; μ: mass per unit length [kg/m]; B: bending rigidity [N·m²].

If diameters D and d are used, $I = \pi(D^4 - d^4)/64$. The radius form and diameter form must never be mixed. For a nonuniform section, compute $I = \int_A y^2\, dA$ about the selected neutral axis. A stiffness-weighted section uses $B = \int_A E(x,y)y^2\, dA$, with the elastic neutral axis satisfying $\int_A Ey\, dA = 0$; density belongs in the mass integral $\mu = \int_A \rho\, dA$ [D1].

Conditional elastic stability

$$P_{cr} = \pi^2 EI / (K_{eff}L)^2$$

Euler buckling applies to an ideal slender, initially straight, elastic column under concentric compression, with the end restraint represented by K_{eff}. Boundary conditions materially affect the critical load [R2]. This model does not resolve crushing, eccentric loading, local wall failure, splitting, creep or fracture.

Material properties and removal law

Wood has direction-dependent properties and variability associated with grain, moisture and growth features [R1]. The single modulus used here is a chosen model coefficient. Represent voids geometrically; reduce E additionally only when a separate constitutive change is justified. Otherwise the same damage can be counted twice.

$$dm_{solid}/dt = -\int_\Omega r_{ext}(x,t)\, dV$$

The removal rate density r_{ext} has units kg/(m³·s). It requires a calibrated law or explicit synthetic input. Hollowing alone specifies no logarithmic or exponential time course. The section formulas give capacity at a geometry; they do not determine when that geometry occurs.

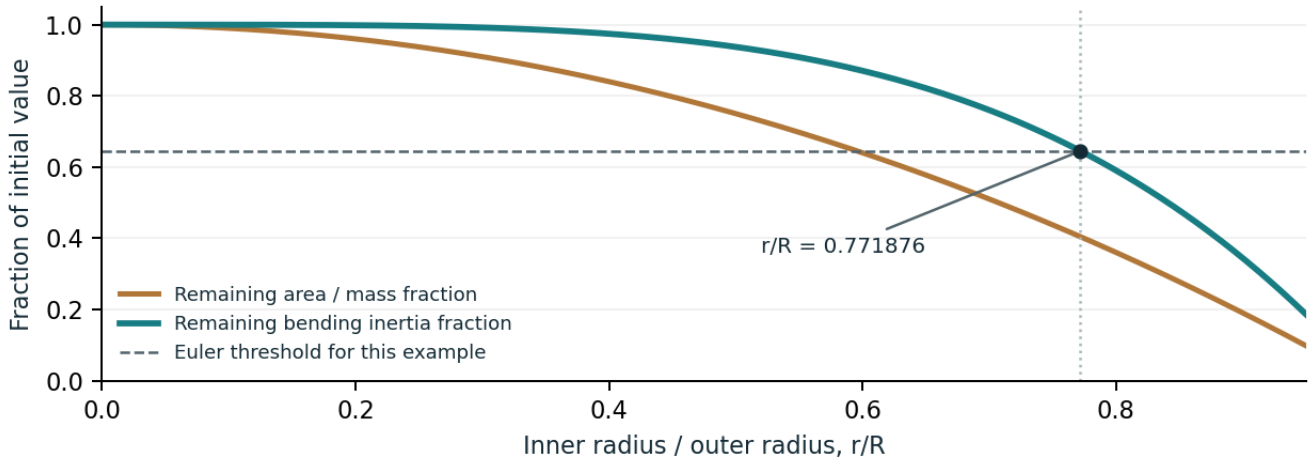
07 / CALCULATED ILLUSTRATION

Worked example: a hollowing column

Synthetic parameters: R = 10.0 mm; L = 1.000 m; E = 10.0 GPa; K_{eff} = 1 (pinned ends); P = 500 N. Hold the remaining modulus fixed and ignore imperfections and other failure modes. These are chosen inputs, not measurements.

$I_0 = 7.853981634 \times 10^{-9}\ m^4$; $P_{cr,0} = 775.156917$ N
$\alpha = r/R$; $A/A_0 = 1 - \alpha^2$; $I/I_0 = 1 - \alpha^4$

r/R	r (mm)	A/A₀	I/I₀	Pcr (N)	Pcr/P
0.00	0.0	1.0000	1.000000	775.157	1.5503
0.50	5.0	0.7500	0.937500	726.710	1.4534
0.75	7.5	0.4375	0.683594	529.892	1.0598
0.80	8.0	0.3600	0.590400	457.653	0.9153
0.90	9.0	0.1900	0.343900	266.576	0.5332



$r_{crit}/R = (1 - P/P_{cr,0})^{1/4} = 0.771876292$
$r_{crit} = 7.7187629$ mm

The threshold formula here requires $0 < P < P_{cr,0}$. At half the outer radius, 25% of cross-sectional area (and mass at fixed length/density) is removed, while bending inertia falls by 6.25%. The nonlinear geometry explains the different curves. No removal rate has been supplied, so no collapse time is asserted. Values and plot are calculated directly from the equations on page 7 [D1].

08 / SPATIAL MECHANICS

Finite elements and stability

For a two-node Euler-Bernoulli element of length ℓ, use nodal coordinates $u_e = (w_1, \varphi_1, w_2, \varphi_2)^T$, with $\varphi = \partial w/\partial z$. Cubic Hermite interpolation supplies displacement and slope continuity [R3]. With constant EI and μ within each element, the usual element matrices are:

$$K_e = (EI/\ell^3) \begin{bmatrix} 12 & 6\ell & -12 & 6\ell \\ 6\ell & 4\ell^2 & -6\ell & 2\ell^2 \\ -12 & -6\ell & 12 & -6\ell \\ 6\ell & 2\ell^2 & -6\ell & 4\ell^2 \end{bmatrix}$$

$$K_{G,e} = (P/30\ell) \begin{bmatrix} 36 & 3\ell & -36 & 3\ell \\ 3\ell & 4\ell^2 & -3\ell & -\ell^2 \\ -36 & -3\ell & 36 & -3\ell \\ 3\ell & -\ell^2 & -3\ell & 4\ell^2 \end{bmatrix}$$

$$M_e = (\mu\ell/420) \begin{bmatrix} 156 & 22\ell & 54 & -13\ell \\ 22\ell & 4\ell^2 & 13\ell & -3\ell^2 \\ 54 & 13\ell & 156 & -22\ell \\ -13\ell & -3\ell^2 & -22\ell & 4\ell^2 \end{bmatrix}$$

$P > 0$ denotes compressive force, so K_G is subtracted from the elastic stiffness. These matrices also follow by integrating $N''^T EI\, N''$, $N'^T P\, N'$ and $N^T \mu N$ over the element; this edition checks them by numerical quadrature [D1].

$$M\ddot{U} + C_d\dot{U} + (K - K_G)U = f$$
$$(K - \lambda K_G(P_{ref}))\varphi = 0;\ P_{cr} = \lambda_{min,+} P_{ref}$$

Apply constraints before testing stability

Assemble the matrices, impose boundary conditions and remove inadmissible rigid modes. For this conservative linear model, positive tangent quadratic form $v^T K_t v$ on admissible nonzero perturbations characterizes local stability. A generalized eigenvalue calculation supplies a meaningful load margin. A determinant sign is not a reliable substitute.

The matrices describe the selected beam approximation at a given geometry. A changing-mass dynamic model requires consistent transport/momentum treatment. Post-buckling motion and fragmentation require additional models and state-transfer rules; linear buckling does not calculate that motion.

09 / PHYSICAL BOUNDARY

Mass, chemistry and heat

A closed system exchanges no mass across its boundary. It may exchange heat and work. An isolated system additionally exchanges no energy [R4]. For the initial model, choose a closed total inventory containing the column, removal agents, fragments, reaction products and surrounding medium. Material removed from the column is transferred internally.

$$dM_{total}/dt = 0$$
$$dE_{total}/dt = Q_{boundary} - \dot{W}_{boundary}$$
$$E_{total} = U_{chemical+thermal+other} + E_{kinetic} + E_{potential}$$

If the loading apparatus is outside the boundary, its work must be included in $\dot{W}_{boundary}$. If the model is declared isolated, boundary heat and work both vanish. Computational access to a state variable is an abstract read; a modeled physical measurement has its own interaction and energy accounting.

Chemical conversion is a configurable submodel

$$(C_6H_{10}O_5)_n + nH_2O \rightarrow nC_6H_{12}O_6$$

This is balanced repeat-unit bookkeeping for cellulose hydrolysis, ignoring polymer end effects. It does not define a universal reaction rate or a complete metabolism. An enzyme or removal law needs concentrations, temperature, moisture and other relevant parameters. In the worked column example, geometry is supplied directly; no biological rate is fitted.

Dimensionally consistent thermal coupling

$$P_{diss} = \mu_f F_n v_{slip} + \int_\Omega \beta \tau \dot{\gamma}_{inelastic}\, dV$$
$$C_{eff} dT/dt = P_{diss} + P_{chem\rightarrow heat} - P_{heat,out}$$

μ_f is friction coefficient, F_n force [N], v_{slip} speed [m/s], τ stress [Pa], $\dot{\gamma}$ strain rate [1/s], and β a heat-conversion fraction. The first term is power [W]; the stress-rate product is power per volume and therefore needs the volume integral. Use both only for distinct processes to avoid double-counting.

C_{eff} is effective heat capacity [J/K]. The lumped temperature equation requires a justified spatial averaging; add transport terms for exchanging subregions. A temperature spike before failure is not implied by material loss. Heat generation, storage and transfer must be calculated together [D1].

10 / GLASS AS INSPECTION

Observation, evidence and entropy

The observation plane distinguishes the modeled state x from what a sensor or observer can infer. This is a technical distinction about data availability. Each record identifies its origin, validity period, uncertainty and transformation history.

$$y_k = h(x_k, c_{sensor}) + \varepsilon_k$$
$$b_{k+1} = \text{Update}(b_k, y_k, \text{measurement_model})$$

h is an observation model and ε a declared noise/error representation. A deterministic estimator can produce the same posterior from the same inputs while the posterior remains uncertain. Missing evidence is represented by a missing-data state, not by a zero value.

Evidence class	Meaning	Required label
Measured	Recorded by an instrument or imported from a named source.	Source, timestamp, units, calibration and quality.
Estimated	Inferred from observations under a model.	Method, assumptions, uncertainty and input references.
Simulated	Produced by the configured state-transition and physical models.	Run, epoch, parameter set and model version.
Illustrative	Chosen to explain a mechanism or test the implementation.	Explicit synthetic/example status.

Two different entropy quantities

$$H(X) = -\Sigma_i\, p_i \log_2 p_i \text{ [bits]}$$
$$S_{thermal} \text{ [J/K]};\ \Delta S_{isolated} \geq 0$$

Shannon entropy measures uncertainty in a probability distribution [R5]. Thermodynamic entropy belongs to a specified physical state and boundary [R4]. There is no general inverse law that makes wood damage increase one while reducing the other.

A NUMERICAL CHECK

For a binary belief, $H(0.5) = 1$ bit and $H(0.99) \approx 0.080793$ bit. A nearly certain belief already has low entropy even if it is wrong. $H = 0$ records certainty in the distribution; it does not independently certify the truth of the belief.

The view must show uncertainty alongside the rendered state. Changes in visibility may inform an observer, but a rendered fracture is not automatically a measurement of real fracture, and an attractive image is not model validation.

11 / RENDERING CONTRACT

Optics and the Glass view

The view maps a committed geometry and material description to light transport. Geometry supplies positions, normals and visibility. Optical parameters supply refractive index, roughness and absorption. Physical status remains a separate categorical field.

Interface reflection

$$F_0 = ((n_i - n_t)/(n_i + n_t))^2$$
$$F(c) \approx F_0 + (1 - F_0)(1 - c)^5$$

The first equation is normal-incidence reflectance at a nonabsorbing dielectric interface [R6]. The second is Schlick’s approximation [R8]. For an illustrative air-to-n=1.5 interface, $F_0 = 0.04$. In a microfacet reflection evaluation, c = clamp(v·h, 0, 1), where h is the normalized half-vector. A material’s measured parameters, not its regime label, determine its optical model.

A full scattering model

$$f_{r,spec}(l,v) = D(h)G(l,v)F(v\cdot h) / [4(n\cdot l)(n\cdot v)]$$

This common microfacet term includes a normal distribution D and visibility/masking term G, for directions in the valid reflecting hemisphere [R7]. Wood generally also needs a diffuse or subsurface contribution. A glass-like material needs reflection and transmission, including refraction and total internal reflection where applicable [R6].

Input	Output behavior	Required check
Updated geometry	Recompute normals and visibility from the same snapshot.	Consistent coordinate frame; valid normal vectors.
Material parameter change	Update the declared BSDF/BRDF parameters.	Valid domain and provenance; no arbitrary regime-to-index jump.
Fracture or exposed surface	Select geometry and surface materials through a supported model.	State transfer and material mapping are explicit.
Diagnostic coloring	Overlay regime, margin or error on the view.	Label the overlay as diagnostics; keep it separate from physical reflectance.

Evaluate the chosen lighting model in its declared linear color space and make exposure and display mapping explicit. Handle degenerate half-vectors and grazing denominators according to the BSDF implementation. A change in normal or roughness does not by itself require a change in refractive index.

12 / ENCODING

Log-polar sampling and GPU LUTs

Use a regularized logarithmic radial map so the center has a finite coordinate. For $0 \le r \le R$ and chosen scale $r_0 > 0$, define:

$$a = \ln(1 + R/r_0)$$
$$u = \ln(1 + r/r_0)/a;\ v = (\theta \bmod 2\pi)/(2\pi)$$
$$r = r_0(\exp(au) - 1);\ \theta = 2\pi v$$

The mapping is invertible away from the polar angle ambiguity at $r = 0$. Give the center one canonical angular value or a dedicated center sample. Define wrapping at the seam and the behavior at $u = 1$. For area integration in (u,v), include the Jacobian:

$$dA = 2\pi\, r\, r_0\, a \exp(au)\, du\, dv$$

This is a sampling-coordinate transformation, not an automatic compression algorithm. It redistributes radial resolution. Interpolation error must be checked in the physical domain after the coordinate transform [D1].

Separate continuous fields from discrete state

Resource	Example contents	Access policy
Numeric field LUT	Strain, displacement, temperature or optical parameters; units in schema.	Interpolation only when physically meaningful within one valid region.
Integer regime resource	State code, material ID, flags, selected behavior ID.	Checked integer indexing or unfiltered integer texel fetch.
Validity resource	Missing-data mask, domain flag, confidence or approximation bound.	Never treat missing values as ordinary interpolatable samples.

OpenGL ES texelFetch uses integer texel coordinates without ordinary filtering or wrapping [R12]. Validate bounds before access. Interpolating a categorical state stored in a floating alpha channel can create a nonexistent regime.

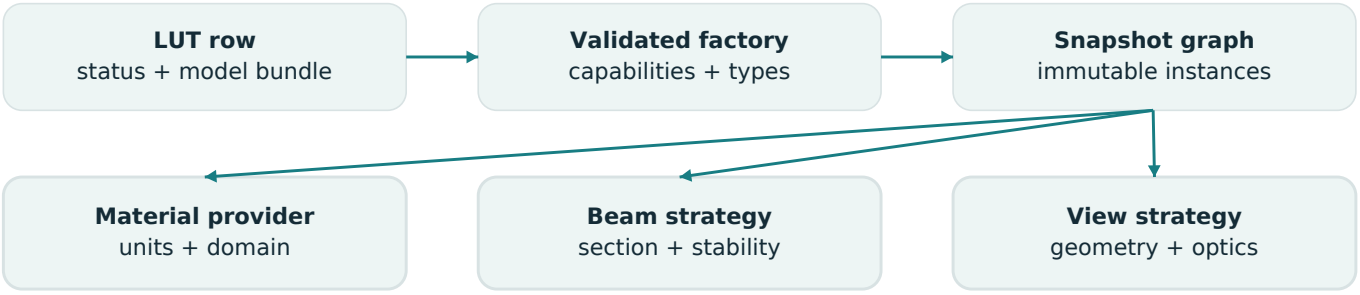
```
continuous = sample_numeric_field(valid_uv, field_policy)
ij         = checked_integer_index(event_or_cell, state_extent)
state_code = fetch_integer_state(ij)
descriptor = checked_behavior_lut(state_code, model_version)
// State IDs are never reconstructed from interpolated colors.
```

LUT data used by a GPU carries the snapshot epoch. CPU publication, GPU transfer, resource barriers and completion fences are separate coordination steps; sharing memory does not remove the need for those contracts.

13 / BEHAVIOR MODEL

Runtime graph and patterns

The source asks for the inheritance diagram to form at runtime from LUT(sS). This specification realizes that requirement as a graph of instances, capabilities and behavior bindings assembled from typed descriptors. In C++, base-class relationships belong to type definitions; runtime selection can choose defined strategies or interfaces [R9].



Top: assembly flow. Branches: retained components. Connections use typed contracts and registered behavior IDs.

```
// Minimal LUT(sS) bundle; all names are registered behavior IDs.
0x00 -> {section: solid,    advance: elastic, view: material}
0x01 -> {section: annulus,  advance: elastic, view: material}
0x02 -> {section: boundary, advance: none,    view: diagnostic}
0x03 -> {section: recorded, advance: none,    view: diagnostic}
```

Selected pattern	Role in this architecture	Boundary
Abstract Factory	Assembles a compatible family of registered components.	Reject mismatched ports, versions or capabilities.
Strategy	Selects mechanics, integration or rendering algorithms.	Each strategy declares a validity domain.
State	Encodes ordered guards and allowed regime transitions.	Runtime fault and material regime remain distinct.
Observer	Delivers versioned change events to downstream consumers.	Ordering and queue overflow behavior are explicit.
Prototype	Instantiates a configuration from a validated template.	Copying shares only immutable resources or creates owned copies.

These are the five patterns selected in [U1], not a universal standard named “Big 5.” In the minimal bundle, critical/failed states support inspection and pause. Continuing motion requires an additional registered strategy [D1].

RESOLUTION OF THE OOB AMBIGUITY

The source user wrote OOB without expanding it. Earlier assistant text expanded it to out-of-bounds and proposed vtable patching. This edition uses bounded composition, with candidate graphs prepared off the active read path. It makes no memory-safety exception for an OOB label.

14 / OWNERSHIP AND LIFETIME

Concurrency: one coherent publication

Use one serialized coordinator and multiple readers. The coordinator prepares and validates a complete candidate off the active read path, then publishes one root reference. Each reader retains that snapshot for its operation. Every referenced descendant must remain immutable while published.

```
// C++20 architecture sketch; domain types are specified separately.
#include <atomic>
#include <memory>
#include <utility>

struct Snapshot { /* state, graph, tables, manifest, epoch */ };
void validate(const Snapshot&);
void evaluate_from(const Snapshot&);

std::atomic<std::shared_ptr<const Snapshot>> current;

void publish(Snapshot next) {             // serialized writer
    validate(next);                       // throws/rejects on failure
    auto owned = std::make_shared<const Snapshot>(std::move(next));
    current.store(std::move(owned), std::memory_order_release);
}

void evaluate() {
    auto snap = current.load(std::memory_order_acquire);
    if (!snap) return;                    // initialization incomplete
    evaluate_from(*snap);                 // owned for this operation
}
```

The atomic shared-pointer facility provides atomic publication and reference ownership [R10]. Its lock-free behavior is implementation-dependent. This sketch does not claim hard real-time latency, and a const outer object alone does not guarantee deep immutability. The implementation must enforce it for graphs, tables and resources.

Publication and reclamation are different operations

If RCU replaces reference ownership, an old snapshot is reclaimed only after the required grace period and completion of its readers [R11]. An atomic raw-pointer swap alone cannot establish that the old data is unused. Double buffering likewise requires proof that no reader or GPU command still uses a buffer before it is overwritten.

Concern	Contract
Stale work	Worker results carry input epoch and model version. The coordinator rejects stale results unless a declared merge rule applies.
Event delivery	Notifications carry commit sequence. Critical events are acknowledged or backpressured; display frames may be coalesced by policy.
Lifetime and budget	Retained snapshots count against memory limits. Slow readers cause backpressure or controlled pausing, never early reclamation.
GPU synchronization	A snapshot’s resource lifetime extends through GPU completion. The graphics API defines transfer and visibility synchronization.

15 / TYPED MECHANICS

Operator catalogue

Operators are typed transformations with explicit error paths. The catalogue replaces metaphorical operator names with contracts that can be implemented and checked [D1].

Operator	Signature / responsibility	Essential condition
Validate	Event × schema → AcceptedInput | Error	Units, bounds, time/order and provenance verified.
Order	Accepted events → deterministic sequence	Tie-break and late-event policy are recorded.
Transfer mass	Inventories × removal law × Δt → inventories	Total closed-system mass conserved within tolerance.
Section	Geometry × material → A, I, B, μ	Radii/diameters distinct; neutral axis defined.
Discretize	Geometry × mesh policy → FEM operators	Boundary conditions and element validity declared.
Integrate	State × operators × input × Δt → candidate	Step-size control, event localization and solver residuals.
Stability	Constrained tangent model → margin	Use a suitable eigenvalue or stability problem.
Heat	Thermal state × power balance → candidate T	Energy and units consistent; no duplicated heat source.
Classify	Regime × candidate × event → regime	Ordered guards; uncertainty and terminal rules explicit.
Lookup	Typed key × LUT version → descriptor | Error	Bounds checked; no silent nominal fallback.
Assemble	Descriptor × registry → graph | Error	Compatible typed ports; bounded graph and resources.
Migrate	Old model state × target contract → target state	Conserved variables and losses accounted for.
Publish	Validated candidate → committed snapshot	Single coherent epoch; ownership established.
Observe	Evidence × measurement model → belief/estimate	Source and uncertainty retained.
Encode	Physical coordinates → sampling coordinates	Finite origin convention; Jacobian for integration.
Render	Snapshot × camera/light model → image	Optical data and diagnostics remain distinguishable.
Journal	Accepted event / decision → replay record	Versions, ordering, rejection reasons and state digest.
Retire	Obsolete snapshot × ownership proof → release	No active CPU reader or GPU use remains.

A stop or pause changes the scheduler state and records why. It does not erase mass, force a physical state to zero, detach live worker resources, or skip the ownership protocol. Shutdown follows the same lifetime rules as ordinary updates.

16 / EXACT MEANINGS

Determinism and automorphism

Determinism is a replay contract

For a fixed initial snapshot, ordered event journal, configuration bundle and numerical execution profile, the reducer produces the same accepted state sequence. Relevant external nondeterminism must be recorded or controlled; record/replay systems demonstrate this principle [R14].

Contract	What must be fixed	How to assess it
Logical replay	Event order, table versions, model IDs, branch policy and accepted substeps.	Compare regimes, decisions and canonical state fields.
Bitwise replay	In addition: arithmetic format, rounding, operation order, fused operations, compiler and target profile.	Compare canonical serialized state digests on the stated profile.
Cross-platform agreement	A documented numerical tolerance and equivalent model configuration.	Compare errors in a declared norm; do not assume identical bits.

Floating-point addition is not generally associative, and fused operations can round differently from separate operations [R13]. Fix authoritative reduction order. Record clock samples, random inputs, network responses, ordering decisions and configuration changes if they can affect results. Parallel workers may run independently, but commit order remains governed by the coordinator [D1].

Automorphic is a symmetry requirement

$$g : S \to S \text{ is bijective}$$
$$g(T_C(s,e,\Delta t)) = T_C(g(s), g_E(e), \Delta t)$$

$T_C(s,e,\Delta t) = s'$ denotes a successful transition with $\text{Step}_C(s,e,\Delta t) = \text{Success}(s')$. The symmetry must preserve that valid transition domain.

On the valid state/event domain, an automorphism preserves the specified transition structure under an invertible relabeling g and matching input transformation g_E. The equation is a property to prove or test. Ordinary feedback, repeated LUT lookup and self-updating state do not establish it.

A MEANINGFUL SYMMETRY EXAMPLE

For an ideal rotationally symmetric column, simultaneous rotation of the spatial fields and inputs can preserve the model if the material, loads and boundaries respect that symmetry. On a discrete polar grid, exact cyclic index shifts are the natural candidates. Arbitrary interpolated rotations require an approximation contract.

Irreversible removal, clamping and quantization are generally not bijections. A system with irreversible evolution can still have spatial symmetries; its time-step operator need not itself be an automorphism. “Automorphic” is therefore reserved for a declared structure-preserving map, while ordinary iteration is called recurrence or feedback [D1].

17 / FINITE IMPLEMENTATION

Accuracy, LUT cost and resource limits

A LUT can exactly encode a finite transition relation or stored bit pattern. A finite numeric table representing a continuous field is an approximation unless the field and interpolation scheme make it exact on the stated domain. Physical accuracy additionally depends on inputs, constitutive models and calibration.

$$N_{entries} = \prod_{i=1..d} n_i$$
$$M_{payload} = b_{entry} \prod_{i=1..d} n_i$$

Here n_i is the number of bins on axis i and b_{entry} is bytes per entry. A checked direct array index has constant address-work for fixed-width keys. Graph construction, numerical integration, matrix solution, allocation and memory traffic remain separate costs. Multilinear interpolation uses 2^d corner samples in d dimensions [D1].

Illustrative resource	Payload
256 × 512 × RGBA32F (16 bytes per texel)	2,097,152 bytes = 2 MiB
256 × 512 × R8UI state resource (1 byte per texel)	131,072 bytes = 0.125 MiB
One copy of both resources	2.125 MiB
Two complete copies	4.25 MiB

These values exclude alignment, mip levels, geometry, graphs, staging buffers, drivers and additional retained snapshots. They are a storage calculation, not a device benchmark.

Declare accuracy and overload policies

Budget or error source	Required declaration
Model and calibration	Validity domain, measured parameters and uncertainty; what the chosen model omits.
Discretization and LUTs	Mesh/time resolution, interpolation policy, precision and an assessed approximation error.
Memory	Maximum table payload, graph nodes, queue depth and retained snapshots.
Update deadline	Measured execution profile and explicit policy for overrun: backpressure, reduced optional rendering, or pause.
History	Bounded active state, journal retention and checkpoint/replay policy.

Reducing optional rendering must not silently change authoritative physics. Changes to simulation fidelity or numerical policy are versioned events. A deadline miss and a memory limit are observable outcomes, not reasons to claim unlimited context or constant-time computation for the whole system.

18 / OPERATIONAL SPECIFICATION

One tick and a replayable trace

The following pseudocode specifies the publication boundary. Model-specific functions must satisfy the contracts on earlier pages; this is not a claim of a deployed runtime.

```
process(requested_event, requested_interval):
    old = acquire_current_snapshot()
    accepted = validate_and_order(requested_event)
    journal_input_or_rejection(accepted)
    if rejected(accepted): return Error(reason)

    dt = choose_valid_substep(old, accepted, requested_interval)
    candidate_x = advance(old.graph, old.state, accepted, dt)
    localize_regime_boundary_or_reject(candidate_x, dt)
    candidate_b = update_observation(old.belief, accepted, candidate_x)
    regime = classify_with_priority(old.regime, candidate_x, accepted)
    desc = checked_lookup(regime, capabilities, bundle_version)
    target_x, target_b = migrate_if_needed(candidate_x, candidate_b, desc)
    graph = assemble_registered_components(desc)
    next = snapshot(old.epoch + 1, old.time + dt,
                    regime, target_x, target_b, graph, manifest)
    validate_invariants_and_budget(next)
    journal_commit_decision(next)
    publish_atomically(next)
    notify_consumers(next.epoch)
    return Success(next)
```

Failure before publication records a rejection and preserves the current physical snapshot. Commit records and recovery must be designed so a restart can distinguish a prepared decision from a published one. If a boundary is localized, dt and candidate state are updated together before classification.

Geometry-controlled demonstration

Row	Requested inner radius	Accepted result	Regime
0	0 mm	Pcr = 775.157 N	STABLE
1	5.0 mm	Pcr = 726.710 N	DEGRADING
2	7.5 mm	Pcr = 529.892 N	DEGRADING
3	8.0 mm along a continuous synthetic path	Localize at r = 7.7187629 mm; Pcr = 500 N. Commit boundary state.	CRITICAL
4	Continue beyond boundary	Pause if no post-critical model is registered. Retain boundary snapshot.	CRITICAL; runtime paused

This trace uses page 8 parameters and zero guard tolerance. Trace rows are not commit epochs or seconds; row 4 pauses without a new physical commit. No removal rate is inferred. A diagnostic critical-state descriptor supports inspection and pausing. FAILED is not entered because the example contains no fracture law or failure observation [D1].

19 / REVIEWABLE COMPLETION

Conformance and implementation sequence

VERIFICATION STATUS

This document recalculates the annular example, checks the displayed FEM matrices against Hermite-basis integrals, checks the binary-entropy values and checks the regularized radial map and inverse. These are arithmetic/model checks. The runtime tests below are acceptance requirements for an implementation.

Check	Required pass condition
Geometry and units	Radius and diameter forms agree; the worked values reproduce to the shown precision. Invalid domains are rejected.
Mass / energy	Closed-model transfer conserves total mass within declared tolerance; energy residual agrees with boundary transfers.
Transition guards	Every valid input has one priority-selected outcome. Unknown keys, missing bounds and unsupported regimes follow explicit policies.
Event localization	The synthetic 7.5-to-8.0 mm step stops at the threshold; no unsupported post-critical step is committed.
Deterministic replay	Same journal and execution profile reproduce the same canonical states. Rejected inputs reproduce the same reasons.
Concurrency / lifetime	Readers never mix epochs; delayed readers and GPU work cannot observe reclaimed or overwritten resources.
Optics / sampling	State IDs remain discrete; the origin and seam are defined; coordinate round-trip and field error stay within tolerance.
Resource exhaustion	Full queues, slow readers and memory/deadline limits invoke the documented policy and leave a valid committed state.
Metadata independence	Changing document author fields does not alter the simulated trajectory or physical parameters.

Implementation order

1. Build a single-threaded reducer with versioned descriptors and the annular example. **2.** Add event journaling, validation and localized transitions. **3.** Add a diagnostic Glass view and checked field/state resources. **4.** Introduce immutable snapshot publication and lifetime tests. **5.** Replace simplified physics with calibrated strategies and verify error bounds.

This order makes the semantics reviewable before concurrency or model complexity obscures them. Device adapters, remote data services and a JIT compiler are optional extensions with separate interface and resource contracts; no such integration is required to define the core paradigm [D1].

20 / TECHNICAL RECONCILIATION

Corrections to the source draft

The source mixes the user's design requests with generated explanations and illustrative execution logs. The formalization preserves the design direction while applying the following corrections. This table is a change record, not a separate theory.

Draft claim or ambiguity in [U1]	Formal treatment in this edition
Identity-derived modulus, wave field or physical load	Personal details remain attribution metadata. Physical parameters require independent definitions.
Feedback automatically means automorphic	Feedback is recurrence. Automorphism requires a bijective, structure-preserving symmetry.
Hollow-section formula uses radii with denominator 64	Use $\pi(R^4-r^4)/4$ for radii, or $\pi(D^4-d^4)/64$ for diameters.
Density inside the area moment used in EI	Geometric I has units m^4. Density enters mass; nonuniform stiffness uses an E-weighted integral.
Hollowing causes exponential/logarithmic stiffness loss	Calculate geometry and material laws. Time dependence needs a specified removal law.
$\det(K-KG) \leq 0$ universally detects failure	Use a constrained stability/eigenvalue problem. Buckling and fracture are separate model events.
Closed and isolated used interchangeably	Closed excludes boundary mass transfer. Isolated additionally excludes boundary energy transfer.
Thermal terms with incompatible dimensions; inevitable heat spike	Integrate power density over volume and balance heat/storage. No universal spike follows.
Structural entropy diverges while information entropy vanishes	Keep thermal entropy and probability entropy distinct. No inverse law is assumed.
Fracture necessarily jumps F0; Fresnel is a complete shader	Geometry, roughness and material changes have separate causes. Use a complete declared scattering model.
Out-of-bounds vtable writes produce safe runtime inheritance	Use registered behaviors and bounded graph composition. Memory barriers do not validate invalid object access.
Pointer swapping / Python example proves lock-free RCU	Publication, ordering and lifetime are separate requirements. No portable lock-free proof is supplied.
O(1) table access means exact unlimited computation	Direct indexing has narrow complexity. Table size, simulation work and active history are bounded.

The earlier "compile successful" and runtime-looking messages are source illustrations. They supply no measurement or build evidence. This edition distinguishes design, calculated example and implementation acceptance explicitly.

21 / REFERENCE KEY

Notation and provenance

Symbol / term	Meaning
C, V	Configuration bundle and its version manifest.
S_k, k, t_k	Committed snapshot, commit sequence and simulated time.
x_k, q_k, b_k	Physical model state, material regime and observation/uncertainty record.
G_k, d, B	Runtime component graph, behavior descriptor and approved behavior registry.
δ, F, Step	Guarded regime transition, physical update and complete runtime transition.
R, r, L; K_eff	Outer radius, inner radius, length; dimensionless end-condition factor.
A, I, E; B = EI	Area [m²], second moment [m⁴], modulus [Pa]; bending rigidity [N·m²].
ρ, μ, P	Density [kg/m³], line mass [kg/m], compression load [N].
K, KG, M, Cd	Elastic stiffness, geometric stiffness, mass and damping matrices.
T, S_thermal, H(X)	Temperature [K], thermodynamic entropy [J/K], Shannon entropy [bits].
n, F0, D, G	Refractive index, normal reflectance, facet distribution and optical visibility; distinguish from runtime symbols by context.
u, v; r0; g	Normalized sampling coordinates; regularization scale; automorphism candidate.

[U1] Supplied source

“Pasted text.txt”, supplied in this conversation. The key source locators are the user prompts beginning “closed system, but not my personal ID” (line 586), “exact and all the operators” (line 713), “inheritance OOB diagram” (line 755), and “multi-thread, design patterns BIG 5” (line 892). Line locators refer to the supplied text copy; short prompt phrases are included so the source remains identifiable if line wrapping changes. The exact source digest is recorded below.

```
Source SHA-256:
2ba9881e54edf7f9f6415f6cf19e8663
d0fc9da4f7d66e570c90e88be25bb97c
```

[D1] This formalization

Original editorial definitions, runtime contracts, diagrams, numerical examples and verification calculations prepared for this edition from [U1]. “Glass” is explicitly a working interpretation; “Googless” is retained as the paradigm name. Cited references support particular established mechanisms, not the novelty or experimental validity of the combined architecture.

22 / SOURCES 1 OF 2

References: physical models

Primary publications and author/institution documentation. Web sources checked 30 October 2026. Section references below identify the relevant scope.

[R1] Senalik, C. A. and Farber, B. (2021). Mechanical Properties of Wood.
Chapter 5, Wood Handbook, USDA Forest Products Laboratory, FPL-GTR-282.

https://research.fs.usda.gov/treesearch/62244

Supports wood-property variability, grain and environmental/material influences.

[R2] Parks, D. M. (2004). Euler-Bernoulli Beams: Bending, Buckling, and Vibration.
MIT 2.002, lecture of 9 February 2004; Euler-column discussion.

https://ocw.mit.edu/courses/2-002-mechanics-and-materials-ii-spring-2004/bc25a56b5a91ad29ca5c7419616686f7_lec2.pdf

Supports the beam approximation and boundary-condition dependence of Euler buckling.

[R3] TU Delft TeachBooks. Euler-Bernoulli beam elements.
Computational Modelling, structural linear analysis, §4.1; Hermite interpolation and stiffness derivation.

https://teachbooks.tudelft.nl/computational-modelling/structural_linear/euler_bernouilli.html

Supports the finite-element construction. Displayed matrices are also checked through their defining integrals [D1].

[R4] MIT Unified Engineering. Concept and Statements of the Second Law.
Thermodynamics and Propulsion, §5.1.

https://web.mit.edu/16.unified/www/FALL/thermodynamics/notes/node37.html

Supports energy/entropy accounting for an isolated system and entropy units.

[R5] Shannon, C. E. (1948). A Mathematical Theory of Communication.
Bell System Technical Journal 27, pp. 379-423 and 623-656; corrected reprint.

https://people.math.harvard.edu/~ctm/home/text/others/shannon/entropy/entropy.pdf

Primary definition of probability-based information entropy and logarithmic units.

[R6] Pharr, M., Jakob, W. and Humphreys, G. Physically Based Rendering, 4th ed.
§9.3, Specular Reflection and Transmission.

https://www.pbr-book.org/4ed/Reflection_Models/Specular_Reflection_and_Transmission

Supports dielectric interface reflection, refraction and transmission distinctions.

[R7] Pharr, M., Jakob, W. and Humphreys, G. Physically Based Rendering, 4th ed.
§9.6, Roughness Using Microfacet Theory.

https://www.pbr-book.org/4ed/Reflection_Models/Roughness_Using_Microfacet_Theory

Supports facet distributions, masking/shadowing and microfacet scattering models.

23 / SOURCES 2 OF 2

References: runtime and computation

[R8] Schlick, C. (1994). An Inexpensive BRDF Model for Physically-based Rendering.
Computer Graphics Forum 13(3), pp. 233-246. DOI: 10.1111/1467-8659.1330233.

https://doi.org/10.1111/1467-8659.1330233

Original rendering model associated with the Fresnel approximation used here.

[R9] C++ working draft. Derived classes.
[class.derived] and [class.derived.general].

https://eel.is/c++draft/class.derived

Supports the distinction between C++ type inheritance and runtime instance/behavior composition.

[R10] C++ working draft. Atomic specialization for shared_ptr.
[util.smartptr.atomic.shared].

https://eel.is/c++draft/util.smartptr.atomic.shared

Supports atomic shared-pointer operations; lock-free support is implementation-dependent.

[R11] Linux kernel documentation. What is RCU? Read, Copy, Update.
Overview, read-side lifetime and grace-period discussion.

https://docs.kernel.org/RCU/whatisRCU.html

Supports separating publication/removal from reclamation after pre-existing readers finish.

[R12] Khronos Group (2016). OpenGL ES 3.1 Specification.
3 November 2016 edition, §11.1.3.2, printed pp. 271-272.

https://registry.khronos.org/OpenGL/specs/es/3.1/es_spec_3.1.pdf

Supports integer-coordinate texelFetch without ordinary filtering/wrapping; indices must be valid.

[R13] NVIDIA. Floating Point and IEEE 754 Compliance for NVIDIA GPUs.
§§2.2-2.3: operation ordering and fused multiply-add.

https://docs.nvidia.com/cuda/floating-point/index.html

Supports numerical-profile requirements; IEEE-format arithmetic alone does not ensure identical results across operation orders.

[R14] rr project. Lightweight recording and deterministic debugging.
Project description of recording and reproducing execution.

https://rr-project.org/

Supports the record/control-nondeterminism principle. The application event schema and reducer contracts in this document are proposed design [D1].